Prerequisites and objectives

- Prerequisites are pointers, object lifetimes, Stack and Queue ADTs, and Chapter 1 invariants.
- Distinguish a node's position, value, and ownership. Identify operations that actually need a search.
- Trace singly linked, circular, and doubly linked lists, linked stacks and queues, and polynomial addition.

Valid singly linked states

- Each node contains a value and next. Head points to the first node. Null represents an empty list.
- Following next from head forms a finite acyclic chain ending in null. Exactly one container is responsible for releasing each node.
- If tail is stored, it points to the last node in a nonempty list and `tail.next=null`. Both head and tail are null for an empty list.
- External pointers may be nonowning references. Deleting the referenced node invalidates them. An address does not imply a live object at that address.

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Insertion with a known predecessor

To insert new node x after p:

1. Allocate and construct x, setting `x.next=p.next`.
2. Set `p.next=x`. If p was the tail, set `tail=x`.
3. Update the element count. If construction fails, preserve the original list.

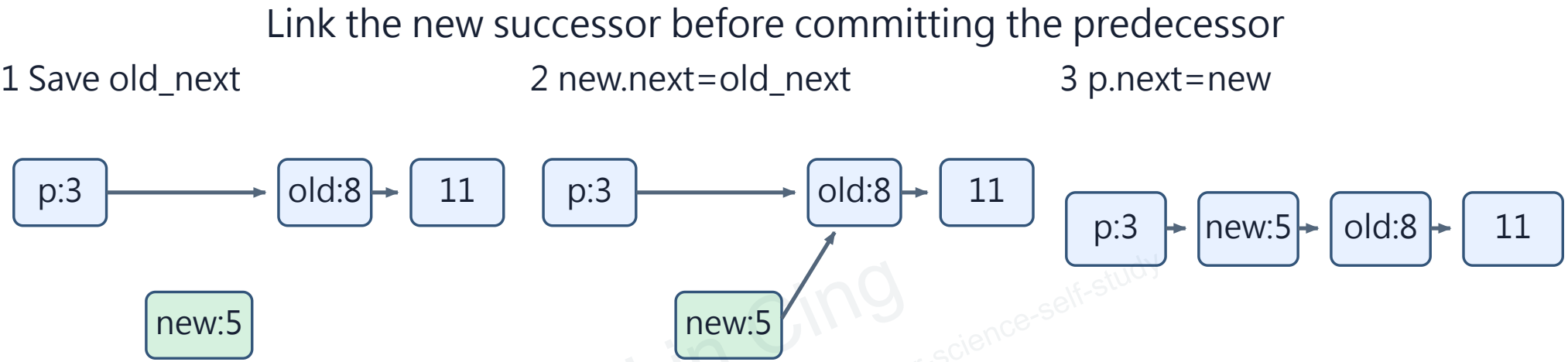
With a valid known predecessor and constant-time allocation, the update costs O(1). Finding the ith predecessor first costs O(i).

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Complete singly linked insertion and erasure

```
insert_after(p,value):
    require p is a live node of this container
    old_next=p.next
    x=allocate_and_construct(value)
    x.next=old_next
    p.next=x
    if tail==p: tail=x
    size+=1
erase_after(p):
    require p is live and p.next!=null
    victim=p.next; after=victim.next
    p.next=after
    if tail==victim: tail=p
    destroy_and_release(victim); size-=1
```

Complete singly linked insertion and erasure (continued)



Insert 5 after p=3 with successor chain 8→11. After allocation x is not reachable from head. Setting x.next=8 preserves the old 3→8→11 path and links x to 8. Committing p.next=x produces 3→5→8→11. Reversing the statements gives x.next=x, losing head reachability of 8 and 11. This is a structural reachability failure.

Complete singly linked insertion and erasure (continued)

Head updates need a separate case or a nondatum dummy predecessor. Expose an empty head/tail state consistently rather than treating the dummy as user data. With a known valid predecessor the link work is O(1); finding that predecessor by position remains O(n). Arbitrary external pointers do not satisfy the ownership precondition.

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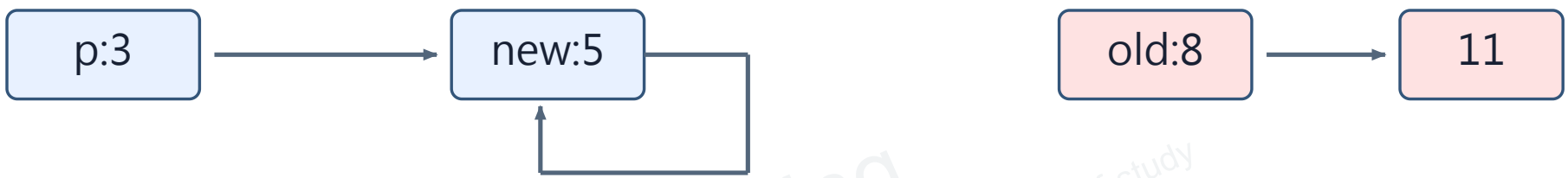
Reachability counterexample for reversed insertion assignments

```
old_next=p.next
p.next=x              // wrong first update
x.next=p.next         // reads x, not old_next
```

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Reachability counterexample for reversed insertion assignments (continued)

Wrong order creates a self-loop and loses the old suffix



p.next=new; new.next=p.next

Unreachable from head

Starting at head=p in3→8→11, the wrong updates produce3→5→5→…. Nodes8,11 remain allocated but lose the owning path from the container. An external pointer retaining their address does not restore the representation. Correct insertion links x to saved old_next before committing p.next=x. The two writes are not automatically atomic for concurrent observers; this chapter assumes single-threaded structural updates.

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Save the successor before deletion

- To delete after p, save `victim=p.next` and `after=victim.next`, set `p.next=after`, then release victim.
- If victim was the tail, set tail=p. Removing the only node requires both head and tail to become null.
- Dereference victim before release. Reading `victim.next` afterward is use-after-free.
- A pointer only to the node being removed does not generally permit O(1) deletion of a singly linked tail. Copying a successor's value also changes node-identity semantics.

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Head operations, queue endpoints, and iterative clearing

```
push_front(v):
    x=Node(v,next=head); head=x
    if tail==null: tail=x
    size+=1
pop_front():
    require head!=null
    x=head; result=copy(x.value); head=x.next
    if head==null: tail=null
    release(x); size-=1; return result
enqueue(v):
```

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Head operations, queue endpoints, and iterative clearing (continued)

```
    x=Node(v,next=null)
    if tail==null: head=x
    else: tail.next=x
    tail=x; size+=1
clear():
    while head!=null:
        x=head; head=x.next; release(x)
    tail=null; size=0
```

These operations share one owning singly linked representation. A linked stack uses push_front/pop_front; a linked queue uses enqueue/pop_front. Assume single-threaded updates and successful allocation/value copying. Copy a possibly throwing result before changing links, so failure preserves the original list. Save the next pointer before releasing a node.

Head operations, queue endpoints, and iterative clearing (continued)

Starting at (head,tail,size)=(null,null,0), enqueue 4 makes both endpoints point to 4. Enqueue 9 gives 4→9 with size 2. Removing 4 leaves head=tail=9 and size 1; removing 9 clears both endpoints and size. Enqueue 7 then creates a valid new chain. Clearing head while retaining the released tail would make that last insertion write through a dangling pointer.

Inductively, insertion adds a fresh node through exactly one owning link, preserves acyclicity, and leaves tail.next=null. Removing the head preserves all remaining nodes and their order. Each update changes size with the reachable-node count. Endpoint operations take worst-case $O(1)$ under fixed-size value and constant allocation costs. Clearing releases each reachable node once in $\Theta(n)$ time and $O(1)$ extra stack space, avoiding a linear recursive-destructor call chain.

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Three-pointer reversal invariant

Initialize `prev=null, curr=head`. Repeatedly save `next=curr.next`, set `curr.next=prev`, then advance prev and curr.

- Prev points to the reversed prefix and curr to the unprocessed suffix. Their node sets are disjoint and their union is the original list.
- After one iteration on `3→8→11`, the reversed part is `3→null` and the unprocessed part is `8→11`.
- At `curr=null`, set head=prev. If tail is maintained, the old head becomes the new tail.
- Every node is handled once. Time is Θ(n) and auxiliary pointer space is Θ(1).

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Complete reversal and the disjoint-segment invariant

```
reverse(head):
    old_head=head; prev=null; curr=head
    while curr!=null:
        after=curr.next; curr.next=prev
        prev=curr; curr=after
    head=prev; tail=old_head
```

Reversing 3→8→11 gives (prev,curr) states (empty,3→8→11),(3,8→11),(8→3,11),(11→8→3,empty). Disjoint segments covering exactly the original node set prove preservation.

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Reversal, one pointer update at a time

Reverse a singly linked list: disjoint parts retain all nodes

Original: 3 → 8 → 11 → null

Step	prev: reversed prefix	curr: unprocessed suffix
0	null	3 → 8 → 11 → null
1	3 → null	8 → 11 → null
2	8 → 3 → null	11 → null
3	11 → 8 → 3 → null	null

Save curr.next before reversing; at the end head=prev and tail=old head.

Empty and single-node chains also work; numbers identify nodes.

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C++ ownership and copying

- `unique_ptr` represents single ownership and transfers the release responsibility with ownership. A nonowning raw tail pointer still needs a validity invariant.
- Default shallow copying of owning raw pointers can make two containers release the same node. Define deep copying or explicitly prohibit copying.
- `shared_ptr` is not an automatic answer for every linked structure. Owning links in both directions or around a cycle create reference-count cycles.
- Recursive destruction of a long chain may create a deep call stack. A teaching implementation can clear iteratively and explain the lifetime behavior.

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Linked stacks and queues

Structure	Pointer arrangement	Update cost
Stack	Head is the top	O(1) insertion and deletion at head
Queue	Remove at head, insert at tail	O(1) each when tail is stored

- Enqueue into an empty linked queue makes both head and tail point to the new node.
- Dequeue of the last node must also set tail to null. Otherwise the next enqueue can write through a dangling pointer.
- These bounds exclude copying arbitrarily large values. An ADT does not guarantee actual allocator latency.

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Canonical polynomial representation

- A node stores `(exponent,coefficient)`. Exponents are nonnegative integers in strictly decreasing order, and coefficients are nonzero.
- $P=4x^5-2x^2+7$ becomes `(5,4)→(2,-2)→(0,7)`.
- Duplicate exponents indicate unnormalized input. Combine their coefficients and remove a node when its coefficient cancels to zero.
- The empty list represents the zero polynomial. Do not mix it with a separate `(0,0)` representation.

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Polynomial addition as sorted merging

- Let $Q=-4x^5+3x^3+2x^2$. Start two cursors at the largest exponents and compare current exponents.
- Output the term with the larger exponent and advance that cursor. For equal exponents, add coefficients, output only a nonzero sum, and advance both.
- The degree-five and degree-two terms cancel, giving $P+Q=3x^3+7$.
- The output remains canonical and equals the sum of consumed input terms. Each node is advanced past once, giving $\Theta(p+q)$ time and $O(p+q)$ output space.

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Polynomial merging with consumed-term states

```
add_polynomials(p,q):
    require both normalized, descending exponents
    out=[]
    while p!=end and q!=end:
        if p.exp>q.exp: append_copy(out,p); advance(p)
        else if q.exp>p.exp: append_copy(out,q); advance(q)
        else:
            c=p.coeff+q.coeff
            if c!=0: append(out,(p.exp,c))
            advance(p); advance(q)
    append_copy(out,remaining(p)); append_copy(out,remaining(q))
    return out
```

For $P=4x^5-2x^2+7$ and $Q=-4x^5+3x^3+2x^2$, equal exponents 5 cancel to zero. Comparing 2 and 3 emits $3x^3$. Equal exponents 2 cancel. Q ends, so append P's remaining 7. The result is $3x^3+7$. At least one cursor advances every iteration, and output remains descending with no zero coefficients.

Polynomial merging with consumed-term states (continued)

Fixed integers require an overflow policy. A floating near-zero tolerance is not exact algebraic equality and changes the abstract value. Choose exact integers, rationals, or an explicitly approximate model before claiming polynomial identities.

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Circular-list termination

- A nonempty circular singly linked list satisfies `tail.next=head`. It does not end at null. Stop traversal on returning to the starting node.
- With tail stored, head is `tail.next`. Inserting at either end changes a constant number of links.
- Starting at B in `A→B→C→A`, one traversal gives B, C, A. Waiting for null would never terminate.
- With a sentinel, an empty list may have the sentinel point to itself. A sentinel is a structural node, not user data.

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Circular tail insertion and endpoint costs

```
circular_push_back(value):
    x=new_node(value)
    if tail==null: x.next=x; tail=x
    else: x.next=tail.next; tail.next=x; tail=x
```

Circular insertion into empty gives A.next=A. Adding B gives tail=B,B.next=A,A.next=B. Traverse with a do/while stopping on return to head, checking tail=null first. With only tail stored, deleting the head is O(1), but deleting the tail needs predecessor search O(n).

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Circular queue: clear tail when its only node leaves

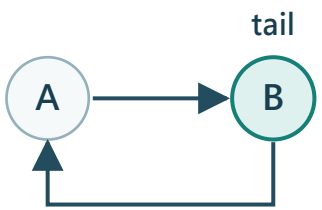
Keep only `tail`; when nonempty, the first node is `tail.next`. If `tail.next==tail`, the ring has one node. Set `tail` to null before releasing that node. Otherwise link `tail.next` to the old first node's successor, then release the old first node.

```
dequeue():
    if tail == null: return empty
    victim = tail.next
    value = copy(victim.value)
    if victim == tail: tail = null
    else: tail.next = victim.next
    destroy_and_release(victim)
    return value
```

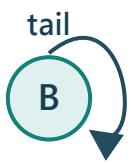
For `A→B→C→A` with tail=C, dequeue yields `B→C→B` with tail still C. For `A→A`, the result must have tail=null. The example assumes successful copying, destruction, and return; potentially throwing generic values need a separate exception contract.

Circular queue: the final dequeue

Circular queue: removing the last node



① Start: A→B→A



② Remove A: B→B

tail = null

③ Remove B: empty

victim=tail.next; for one node, set tail=null before releasing victim.

A later enqueue creates C→C; released B must not be dereferenced.

Highlighted node is tail; the empty queue has no valid tail.

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Floyd's pointers: find a cycle or reach the end

Start both pointers at head. Before each iteration require that fast and fast.next are non-null; advance slow once and fast twice. A meeting implies a cycle. If the check fails, the finite chain reached its end. Nodes must remain live and the structure must not be modified concurrently.

```
slow=head; fast=head
while fast != null and fast.next != null:
    slow=slow.next; fast=fast.next.next
    if slow==fast: return meeting
return no_cycle
```

An empty chain and a single node ending in null have no cycle. A single self-linked node meets on the first iteration. The algorithm never changes links.

Why does the second meeting find the entry?

Let μ be the number of edges from head to the cycle entry and $\lambda \geq 1$ the cycle length. The first meeting is inside the cycle. If slow has taken t steps and fast $2t$, equal positions imply $t \equiv 0 \pmod{\lambda}$.

Reset one pointer to head, leave the other at the meeting, and advance each one step per iteration. After μ steps the first reaches the entry, and the second is at $(t - \mu + \mu) \bmod \lambda = 0$ relative to that entry. Earlier, the head pointer is outside the cycle while the other is inside, so they cannot meet. Their second meeting is exactly the entry.

The cyclic case takes $O(\mu + \lambda)$ time and $O(1)$ extra pointer space. A finite acyclic chain of length n takes $O(n)$. The argument compares node identity and does not assume distinct stored values.

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Trace: a two-edge prefix and a three-node cycle

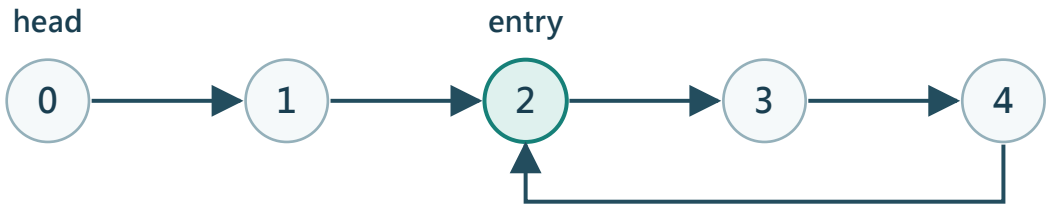
The links are `0→1→2→3→4→2`, so $\mu = 2$ and $\lambda = 3$. Successive `(slow,fast)` pairs before the first meeting are `(1,2)`, `(2,4)`, `(3,3)`. After resetting one pointer, the pairs are `(0,3)`, `(1,4)`, `(2,2)`, locating entry 2.

Compare node identity, not stored value. Locating an entry and breaking a cycle are different operations; this algorithm returns the entry without changing any `next` link.

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Two meetings of Floyd’ s pointers

Floyd: first meeting, then locate the entry



$\mu=2$
$\lambda=3$

First phase: slow takes one step; fast takes two

(1,2) → (2,4) → (3,3)

Reset one pointer to head; advance both one step

(0,3) → (1,4) → (2,2)

Compare node identity; the first meeting need not be the entry.

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Local consistency in doubly linked lists

- Nodes have prev and next. Neighboring nodes satisfy `x.next.prev=x` and `x.prev.next=x`, with endpoints handled by the sentinel policy.
- To insert x between adjacent p and q, set `x.prev=p,x.next=q,p.next=x,q.prev=x`.
- To delete known x, set `x.prev.next=x.next` and `x.next.prev=x.prev`, then release x.
- Updating a known node costs O(1). Finding a node by value still costs O(n). Removing a known node and searching then removing are different operations.

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Doubly linked erasure and reference lifetime

```
doubly_erase(x):
    require x is a live owned data node, x!=H and x!=T
    p=x.prev; q=x.next
    p.next=q; q.prev=p
    release(x); size-=1
```

This procedure fixes two nondata sentinels H and T. H.prev=null and T.next=null; an empty list has H.next=T, T.prev=H, and size=0. Every data node has nonnull prev/next, making p.next and q.prev valid. A null-ended representation instead needs explicit head/tail repair when erasing an endpoint and cannot directly use this procedure.

For one item H↔x↔T, p=H and q=T. Erasure restores H.next=T and T.prev=H and decreases size from 1 to 0. Never release H or T. The same sentinel links cover first/last data nodes in longer lists. Validate the live owned handle before entering pointer operations.

Doubly linked erasure and reference lifetime (continued)

Deleting p↔x↔q sets both p.next=q and q.prev=p. The reciprocal conditions are restored together. References to surviving nodes remain valid; every reference to x is invalid. Updating only one direction gives inconsistent traversal sequences.

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Doubly linked list: removing the only node

Doubly linked sentinels: remove the only data node



① Save p=x.prev=H and q=x.next=T
② Set p.next=q and q.prev=p
③ Release x; size changes from 1 to 0



H.next=T
T.prev=H

H and T remain valid; external pointers to x must not be dereferenced.

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Choosing lists or arrays

Operation or constraint	Dynamic array	Singly linked list
Indexed access	O(1)	O(n)
Insertion with known predecessor	O(n) shifts in the worst case	O(1) link changes
Sequential scan	Usually storage locality	Pointer following
Extra space per item	Unused capacity	Next link and allocation overhead

The complete workload determines the tradeoff. Link counts alone do not establish measured speed.

References and attribution

Lin Cing, handwritten data structures notes; C++ draft forward_list, list and unique_ptr.

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